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Problem Chosen

B**2005 Mathematical Contest in Modeling (MCM) Summary Sheet**

(Attach a copy of this page to each copy of your solution paper.)

Type a summary of your results on this page. Do not include the name of your school, advisor, or team members on this page.

Summary

This paper is aimed to choose the optimal number of the tollbooths in the toll plaza.

The optimal number of the tollbooths is supposed to maximize traffic flow the toll plaza allowed, and to minimize the average time a car passing the plaza, considering the economic reason in addition.

A series-two-stage model is developed based on the queuing theory. We model the process of arrivals and the tolling as the $M/M/K$ problem. And also the process of cars leaving the tollbooths is modeled as another unusual queuing problem, whose service time is determined by the relationship between cars' velocity and the traffic flow density. Those two processes are in series; hence we treat them as a service system of two service stages in series.

The Monte Carlo method is used to simulate system in the situations (such as the different ways of paying and the types of the vehicles) with different traffic arrival rates and different numbers of tollbooths. By simulations, we can get the average time to pass the plaza and the maximum traffic flow the plaza allowed, as the results of setting different number of tollbooths. We choose the optimal number of the tollbooths, when these two factors meet the objectives well.

Finally, we analyze the results of the simulations in detail, explain the traffic congestion, find the bottlenecks of the system, and explicitly analyze the scenario where there is exactly one tollbooth per incoming travel lane.

Is the Number of Tollbooths Optimal

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Introduction

Recently, problems about tollbooths on freeways have attracted considerable public attention. The government builds those tollbooths in two major purposes: to regular the traffic, and to collect tolls. However, it's familiar to us that the toll plazas always interrupt traffic on multi-lane highways. Vehicles have to slow down on entering the tollbooths and squeeze back to a number of lanes equal to the number of lanes entering plaza. Congestion is likely to occur upon the departure from the plaza when traffic is heavy. If the traffic is very heavy, congestion may also occur at the entrance because of the toll-paying time.

Our objective is to find the optimal number of the tollbooths, which is supposed to both maximize the traffic flow the system can support and minimize the time consumed in the system. Based on queuing theory, we design a two-serial-stage model, treating the processes of cars entering the plaza and leaving plaza-outbound as two service stages in series. When vehicles come down to the plaza, drivers are likely to choose the shortest queue ahead waiting for tolling. At the tollgates, three different kinds of methods to pay the tolls are considered. Drivers could pay the tolls either by cash, by tickets or by electronic scanning of ETS; these three methods differ in the time paying tolls. Thus, this stage is treated as the first service stage. After they pay their tolls, they would squeeze to the plaza-outbound lanes, which are much narrower than the plaza. In this case, leaving the plaza is treated as the second service stage. Altogether, these two stages are taken as a two-serial-stage system.

Using Monte-Carlo method, we simulate these two stages and get abundant results. We carefully analyze them in detail, explain the traffic congestion, and find the bottleneck of the system. Specially, we consider the scenario where there is exactly one tollbooth per incoming lane, and compared the efficiencies of this case with the current practice in different conditions.

Assumptions

- The arrival of the cars obeys Poisson distribution, and the service time at the tollbooths obeys the exponential distribution; we will further confirm them in the **Model Design** section below.

- On entering the plaza, drivers pick their routes leading to the shortest queue ahead in front of the tollbooth. Moreover, this route-picking process would not significantly affect the traffic flow behind them.
- Drivers tend to maintain a safe distance from the vehicle ahead of them and there's no tailgating.
- There is not any traffic accident in the toll plaza or on the highways.
- The weather condition is under the state that doesn't affect traffics.

Problem analysis

It's obvious that we can divide the process for a vehicle to pass the toll plaza into three sub-processes: the deceleration to enter the tolling plaza; waiting for its turn to toll in the tolling queue; paying the toll at the tollbooths, squeezing and accelerating to leave the tollbooths and enter the outgoing lane.

Entering process. On entering the plaza, all vehicles need to decelerate in advance while simultaneously drivers choose to append to the shortest queue waiting for paying tolls. Considering the number of tollbooths won't significantly affect the time for a car to decelerate into the plaza, we treat, for simplification, the deceleration time irrelevant to the number of tollbooths we need to optimize.

Tolling process.

- 1) Waiting in queues for paying the toll. If drivers join in a queue not empty, they have to stay in the queue and wait for their turns to pay the toll.
- 2) Paying the toll. There are mainly two methods of paying tolls: by cash/ticket, or just going through the tollgate equipped with an electronic toll system (ETS). They have different time of paying the toll. We still consider different types of vehicles also differ in the time for paying the toll even using the same tolling method.

As in our analysis, the process is a queuing and serving process, which can use queuing theory to model.

Leaving process. After a car having paid the toll, it starts to leave the tollbooth for the outbound lanes. If the number of tollbooths is larger than the number of outbound lanes, there is a chance that many cars from the tollbooths compete one outgoing lane for leaving. It's a process of squeezing flows into a bigger one. Also as we know, cars can only enter the outgoing lane one by one. It will make other cars wait for leaving. In fact, in our further discussion, we can still model the waiting process as a queuing model.

By analyzing these sub-processes, we find the number of tollbooths significantly affects two main processes: tolling process and leaving process. They are modeled as two serial service stages. Figure 1 below shows the brief model of the process of passing the toll plaza.

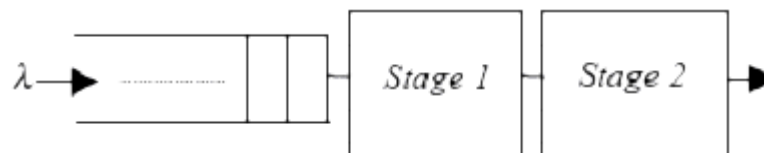


Figure 1. The model of series-two-stage system

Objectives

Higher flow. The toll plazas are generally considered to interrupt the traffic on highways. To reduce the impact of interruption, the optimal number of tollbooths should bear significantly higher traffic flow than others.

Less time. The time required for vehicles to pass the plaza is composed of the time spent in deceleration, waiting for paying the tolls and being severed at the tollbooth, “queuing” for leaving and acceleration to leave. In accordance with the higher flow ideas, this time should be as little as possible. As the whole process is stochastic, all the time is statistical time or average time. When a decision bears a higher flow, traffic would get smoother, and result in less time consequently. Thus, *higher flow* is uniform with *less time* in most cases.

More economical. From the government’s perspective, less tollbooth should be built for the economic reason. Hence if building another tollbooth would not contribute much to the decrease of time, or equivalently, the increase of traffic flow, we are not likely to add this useless tollbooth.

However, the *higher flow* and the *less time* are not consistent with the *more economical*. Thus, we would choose the optimal decision in an idiographic case by comparing each solution with these three objectives.

Terms and Symbols

- *Server* is a tollbooth of the plaza, or a lane of the plaza-outbound lanes.
- *Stage* is the set of servers, which is the combination of the tollbooths, or the plaza-outbound lanes.
- *System* is the serial combination of the two stages; they are the tollbooths and the plaza-outbound lanes.
- *The pass-time* is the span of time that a driver needs to pass the plaza, from deceleration to recovering the normal speed.
- *The sojourn time* is the span of time for a vehicle to stay in a specified service stage.
- *The arrival rate* is the number of vehicles entering the toll-plaza per hour.
- *The safety headway distance* is the space between the back of the leading car and the front of the immediately trailing car.

Frequently used symbols in this paper are listed in the table below.

Table 1. Symbols used in this paper

Symbols		Definition
Time	T_d	The span of time for decelerating into the plaza
	T_a	The span of time for accelerating out of the plaza
	T_{s1}	The sojourn time in the first service stage
	T_{s2}	The sojourn time in the second service stage
	T_E	The span of time for a vehicle to enter the plaza
	T_L	The span of time for a vehicle to leave the plaza

	<i>T</i>	The pass-time of a vehicle
	<i>S</i>	The service time at a service stage
Traffic	<i>n</i>	The number of lanes coming to the tollbooths
	<i>λ</i>	The arrival rate
	<i>l</i>	The safety headway distance
	<i>v</i>	The speed of a vehicle
	<i>F</i>	The traffic flow
Tolling	<i>u</i>	The average time of paying tolls
	<i>k</i>	The number of tollbooths at the plaza

Model design

Entering the plaza:

Vehicles on highways are coming in several lanes, and they can be reasonably treated as a single source of customers. From this single source, vehicles are coming independently. Their arrival won't affect the arrival vehicles behind them, which is the Markov property we called. Among the probability distributions, the exponential distribution is proved to have the Markov property. Thus, we assume that the interval time Δt of two vehicles coming to the plaza obeys the exponential distribution. Hence, the incoming of customers are the Poisson stream.

$$\begin{cases} P(\Delta t) = \lambda e^{-\lambda \Delta t} & \Delta t > 0 \\ P_n(t) = \frac{(\lambda t)^n}{n!} \cdot e^{-\lambda t} & t > 0 \\ n = 0, 1, 2, 3 \Lambda \end{cases}$$

where $P(\Delta t)$ represents the probability of the interval time Δt , $P_n(t)$ represents the probability of n cars arrival the toll plaza in time length t , λ is the average number of the arrivals in a certain time.

Figure 2 shows an example of the probability of Poisson distribution

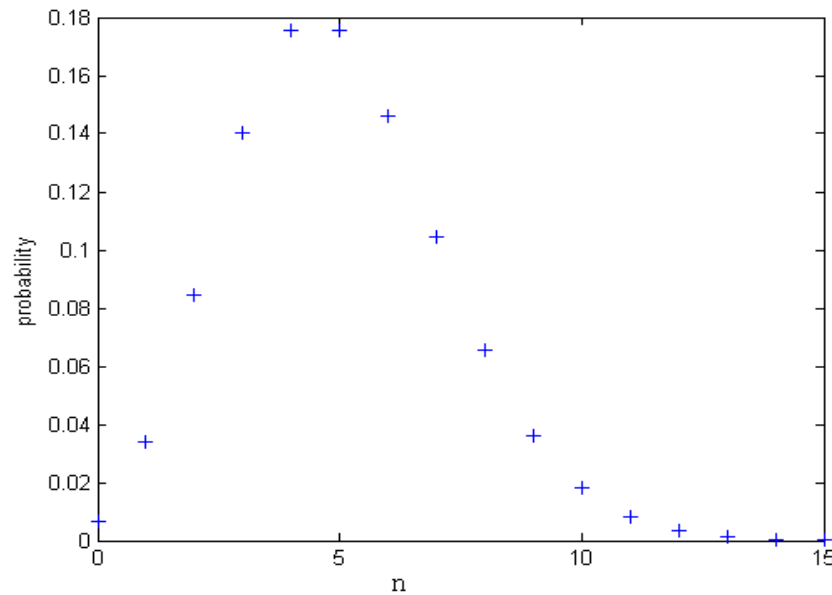


Figure 2. An example of Poisson distribution

Suppose a driver is entering the plaza. We assume that upon entering the plaza, he adjusts his routes to the shortest queue in front of the tollbooths. It is likely to insignificantly affect the traffic flow behind him on the incoming lanes. The span of time since he starts to decelerate until he appends to the queue is called the deceleration time. After he has appended to the shortest queue, he has to wait for service until all vehicles ahead of him are tolled.

It's reasonable to assume that each of the service time for tolling is independent to others and obeys the exponential distribution.

$$P_T(t) = \frac{1}{\mu} \cdot e^{-\frac{t}{\mu}} \quad t \geq 0$$

where $P_T(t)$ represents the probability of the service time for one time tolling of one tollbooth t , μ is the average time for a tollbooth to serve (toll) the one car.

Figure 3 shows an example of the probability of exponential distribution.

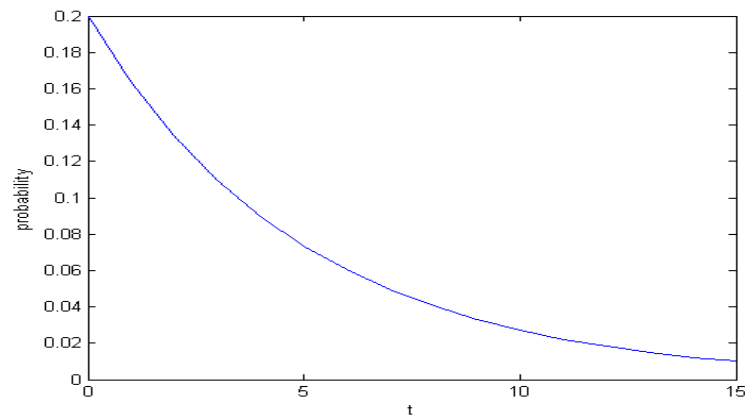


Figure 3. An example of exponential distribution.

If there are k tollbooths in the toll plaza, we can assume that $u_1 = u_2 = \dots = u_k$, by the queuing theory, we can use

the classical **M/M/K model** to model the process of tolling.

In the **M/M/K model**, when in the statistic balance situation, which means the in-flow is less than the capability of tollbooths, that is

$$\rho = \frac{\lambda}{k\mu} < 1.$$

Here ρ is the intensity of service. The probability of the cars staying in the system (Here “staying in the system” means that a car is tolling or waiting for tolling in the queues)

$$P_0 = \left[\sum_{i=0}^{k-1} \frac{1}{i!} \left(\frac{\lambda}{\mu}\right)^i + \frac{1}{k!} \cdot \frac{1}{1 - \frac{\lambda}{k\mu}} \cdot \left(\frac{\lambda}{\mu}\right)^k \right]^{-1}$$

$$P_n = \begin{cases} \frac{1}{n!} \left(\frac{\lambda}{\mu}\right)^n \cdot P_0, & n \leq k, \\ \frac{1}{k! k^{n-k}} \cdot \left(\frac{\lambda}{\mu}\right)^n \cdot P_0 & n > 0. \end{cases}$$

where P_0 means there is no car staying in the system, and P_n means there are n cars in the system.

By these equations, we can compute the following useful indexes of this system:

(1) Average number of cars in each queue N_q :

$$N_q = \sum_{i=k+1}^{\infty} (i - K) P_i = \frac{(K\rho)^K}{K!(1 - \rho)} \rho P_0$$

(2) Average wait time t_q :

$$t_q = \frac{N_q}{\lambda}$$

(3) Average time of staying in system t_{stay}

$$t_{stay} = \frac{N_q + K\rho}{\lambda}$$

We can use these equations to compute these indexes of the tolling server precisely. In another way, we can still use Monte Carlo simulation to compute the results of these indexes. The process for one car to pass the tollbooth is: when there are j cars in the queue in front of the car. The total time for the car to pass the tollbooth can be divided into two parts: the waiting time $t_{1,w}$ in the queue and its service time $t_{1,s}$ in the tollbooth. And in the queuing theory, we know that the waiting time is the cumulative service time of its front cars, which means

$$t_{1,w} = \sum_{i=1}^j t_{1,i},$$

where $t_{1,i}$ means the i th car's service time of tolling. So the whole time T_{s1} for a car to toll is

$$T_{s1} = \sum_{i=1}^j t_{1,i} + t_{1,s}.$$

So finally we can calculate the average whole time for a car to toll, here the average time can be expressed as $\bar{T}_{s1}$.

Leaving the plaza:

After the drivers have paid the tolls, they run out. For example, shown in figure below. The outgoing road has two lanes, and there are k tollbooths in the toll plaza. In the situation $k > 2$, and the running cars accelerate to enter the lanes for leaving the plaza. The cars may squeeze and pipe out to the outgoing lanes. And for the sake of safety, a car can go only there is a safe gap distance L between itself and its front car in the outgoing lane. Here we compute the time Δt the front car accelerates to get out and go the safe distance L

$$\Delta t = \int \frac{dL}{v(t)} = \sqrt{\frac{2L}{a}},$$

where $v(t)$ is the speed of the car running outbound to the lanes.

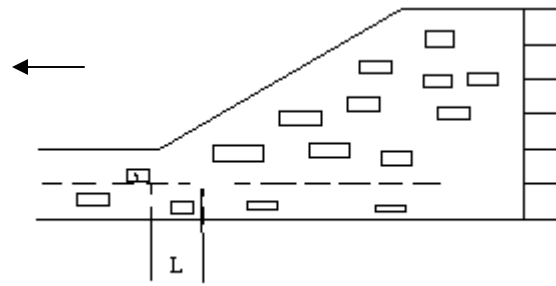


Figure 4. The running cars are squeezing out into outgoing lanes.

But the time one car spent at the tollbooths is stochastic. There is a chance that too many cars leave from the tollbooths; they all contend the lane to get out. But indeed they can just enter into the outgoing lane one by one in order; the waiting cars must stop and queue to wait for their orders to leave. So the process of cars outgoing are just like queuing. Hence we abstract this leaving time as the service time in queuing theory. Figure 5., figure 6 and figure 7 on the next page explain this transformation in graphs.

Also as we know, too many cars will cause the increase of density of traffic flow; it will induce the decrease of outgoing velocity, equation $v = v_0 \cdot (1 - \rho/\rho_0)$ can be used to compute the real velocity (ref [2]), where v_0 represents the speed that a vehicle would travel in the absence of other traffic (the speed limit). ρ is the car density, and ρ_0 is the maximum density, and. When density reaches the maximum the traffic is jammed. And in our abstract queue, $N_2/N_{2\max} = \rho/\rho_0$, N_2 means the number of cars in the queue and $N_{2\max}$ means the capability of the queue. Hence, the service time

$$\begin{aligned}
 t_2 &= \int \frac{dL}{v} = \int \frac{dL}{v_0 \cdot (1 - \rho/\rho_0)} \\
 &= \frac{1}{1 - \frac{\rho}{\rho_0}} \cdot \int \frac{dL}{v_0} \\
 &= \frac{\Delta t}{1 - \frac{N_2}{N_{2\max}}}
 \end{aligned}$$

The decrease of outgoing velocity will cause the increase of the service time in outgoing process. It's reasonable that the coming cars should choose and join in the shortest queue. We can also conclude that the more cars come out from the tollbooths, the more cars will be in the outgoing queues. It may cause congestion.

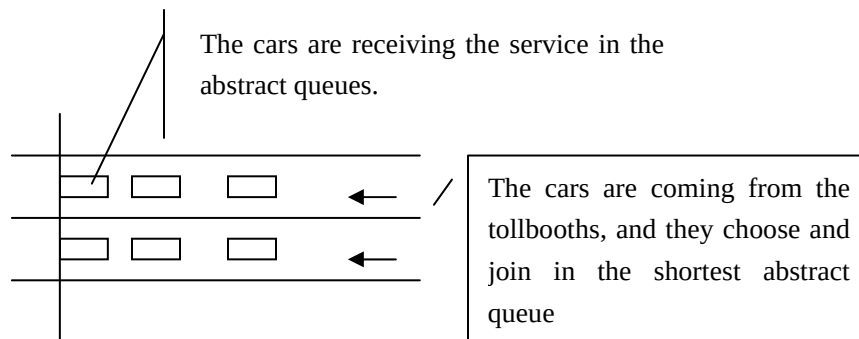


Figure 5. The abstract queues of leaving the plaza

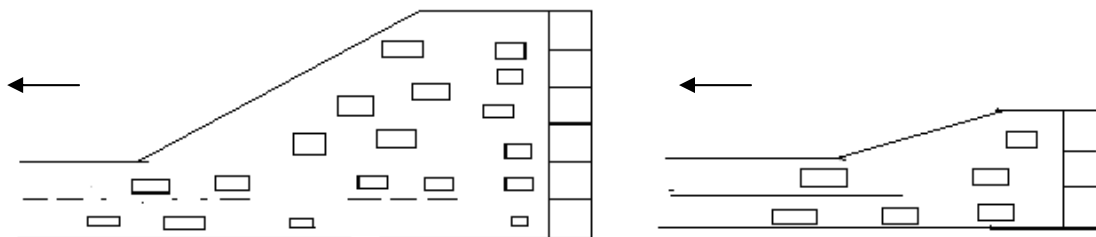


Figure 6. More tollbooths may cause more cars (left);
Fewer tollbooths may cause fewer cars (right)

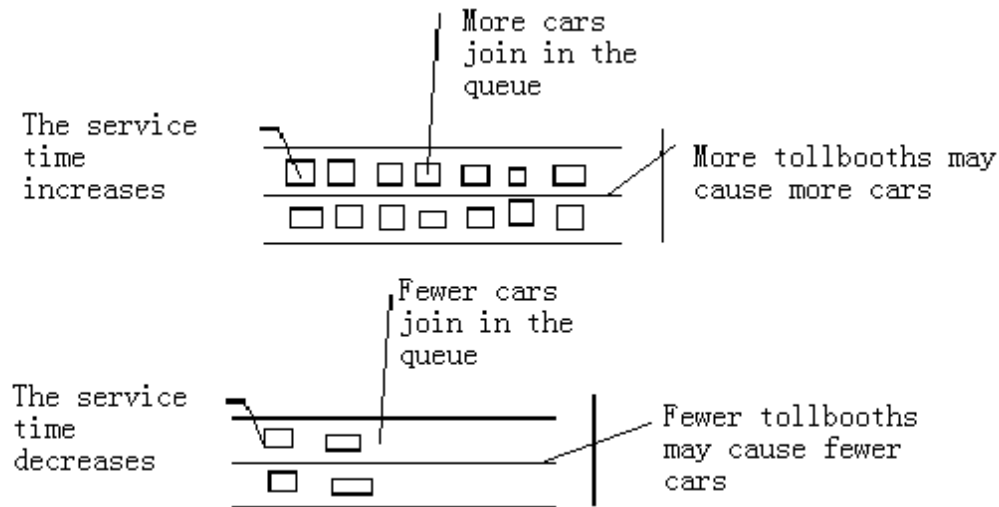


Figure 7. The abstract queues in two situations

In order to calculate the average staying time for one car in this outgoing server, we find it's difficult to compute the analytic answer, so we use the Monte Carlo simulation method to get a numeric statistic answer. It's obvious that for one car, in the particular time when there are j cars in the outgoing queue in front of it. The total time for the car to leave can be divided into two parts: the waiting time $t_{2,w}$ in the queue and its service time $t_{2,s}$. And in the queuing

theory, we know that the waiting time is the cumulative service time of its front cars, which means $t_{2,w} = \sum_{i=1}^j t_{2,i}$, where $t_{2,i}$ means the i th car's service time. So the whole outgoing time T_{s2} for a car to leave is

$$T_{s2} = \sum_{i=1}^j t_{2,i} + t_{2,s}$$

So finally we can calculate the average whole outgoing time for a car to leave and enter the out lane, the average time can be expressed as $\bar{T}_{s2}$.

Model results

Simulating the model

As we see, the model we develop is a complex stochastic service system. It seems very difficult to find the final analytic solution of this problem. So we design a Monte Carlo simulative program to simulate these systems with different parameters (Here the parameters are the arrival rate λ and the number of tollbooths we need to optimize. By

computing the final statistic results of our objective for optimization. In case of different λ, k , the tollbooth system has different performances: the time cost by passing the system may be different, the traffic flow crossing the system may be different, mean number of cars in the system may be different ... So we can choose the number of tollbooths k that can make the system meets our optimal objects.

Generally, there are two kinds of simulation ways for simulation: time-based and events-based ways. Although time-based algorithm is easy to implement, but to reduce the simulation time, we use events-based Monte Carlo algorithm. The flow chart of our simulation algorithm is presented in figure 8 on the next page. Also we provide our Matlab program for simulation in our appendix.

We define three kinds of events:

1. **A car enters the tolling plaza:** it will enter the shortest queue and waiting for tolling.
2. **A car tolls in the tollbooth:** it will be removed from its tolling-queue and then adding into the shortest queue of outgoing lanes.
3. **A car leaves the system:** it will be removed from its queue for outgoing queue.

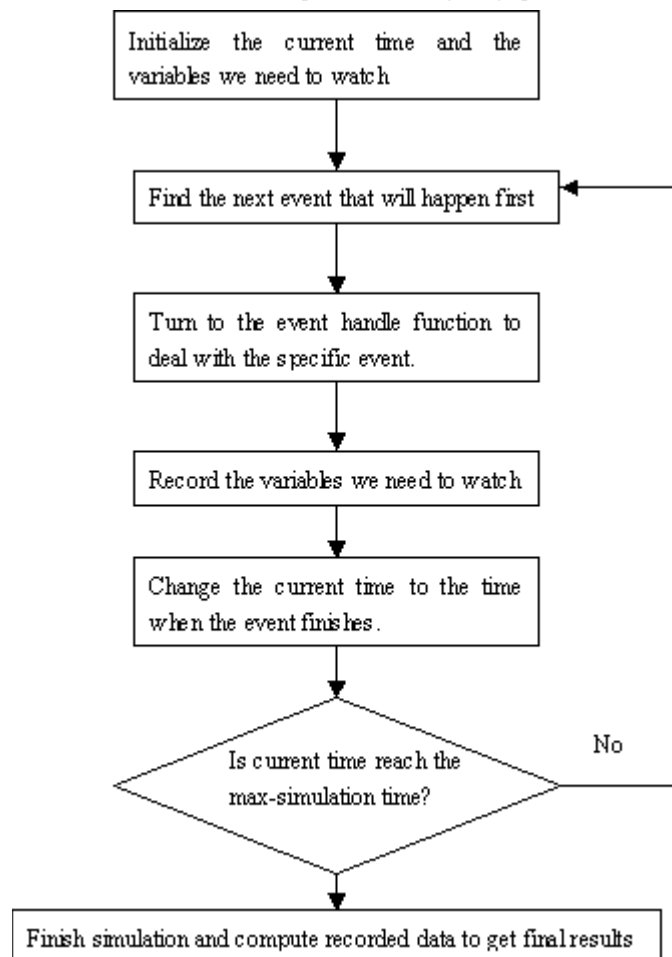


Figure 8. The flow chart of our algorithms

Model computation and analysis

We continue to consider the situation of two incoming lanes. We use our simulation algorithm to simulate the systems with the following situation:

We specially consider the scenario where there is exactly one tollbooth per incoming lane. Thus there are 2

tollbooths in the system.

First we don't consider the effects of different types of vehicles, different paying styles have different time for giving the service, here we consider two popular paying styles: paying by ticket or cash and paying by ETS.

One tollbooth per incoming lane

Here we assume $\bar{L}=50m$, $\bar{a}=2m/s^2$. By our computation, we get the results as listed in table 2 and table 3.

They, respectively, show the results by different service types: tolls paid by cash or ticket, or by passing through the electronic toll system (ETS). As in the **Terms and symbols** section, the variables in the tables are

- λ is the arrival rate;
- $\bar{T}$ is the mean pass-time;
- $\bar{T}_{s1}$ is the mean sojourn time of the first service stage, the tollbooths;
- $\bar{T}_{s2}$ is the mean sojourn time of the second service stage, the plaza-outbound lanes;
- F is the mean traffic flow;
- N_1 is mean number of cars in each tolling queue;
- N_2 is mean number of cars in each leaving queue;

Also, the number in bold and underlined in the tables represents there's a corresponding congestion.

Table 2. Service type: paying by cash or ticket (average service time $\bar{S}=15s$)

λ (cars/hr)	$\bar{T}$ (s)	$\bar{T}_{s1}$ (s)	$\bar{T}_{s2}$ (s)	F (cars/hr)	N_1 (cars)	N_2 (cars)
300	32.59	25.04	7.54	298.5	1.45	0.77
400	59.58	51.67	7.90	400.75	3.31	0.81
500	<u>1186.6</u>	<u>1178.3</u>	8.41	486.3	<u>81.3</u>	0.90

Table 3. Service type: paying by ETS (average service time $\bar{S}=5s$)

λ (cars/hr)	$\bar{T}$ (s)	$\bar{T}_{s1}$ (s)	$\bar{T}_{s2}$ (s)	F (cars/hr)	N_1 (cars)	N_2 (cars)
400	13.69	5.75	7.93	400.8	0.87	0.83
500	14.38	5.97	8.40	493.8	0.90	0.91
600	15.60	6.36	9.24	602.7	0.99	1.07
700	<u>493</u>	7.09	<u>485</u>	620	1.13	<u>97</u>

800	<u>17129</u>	7.64	<u>17121</u>	339.6	1.27	<u>2107</u>
3000	<u>79097</u>	<u>47811</u>	<u>67789</u>	77.8	<u>1984.8</u>	<u>1385.2</u>

By the contrast of the two tables, we can conclude that the two paying styles have different bottlenecks:

In the style paying by cash or ticket, we can clearly see the congestion first appears in the entering the tollbooths, which means the service capability of tollbooths has become the bottleneck of the system. We can solve the problem easily by increase the number of tollbooths.

In the style paying by ETS, congestion first appears out from the toll plaza, which means the squeezing to leave process has become the bottleneck of the system. Simply increasing the number of tollbooths can't solve this problem. In this situation, most of time, one tollbooth per incoming travel lane is always more or less effective than the current practice. We will discuss it clearly in following analysis of our results.

Analyze data to explain the phenomena about traffic and tolling

From table 3, we can see that when the arrival rate increases, it will induce the increase of these results.

As we see, when traffic isn't heavy, mean traffic flow is nearly equal to the arrival flow, and that means the system is in a dynamic balance state; the system can deal with all the incoming cars. Here is the diagram to illustrate it.

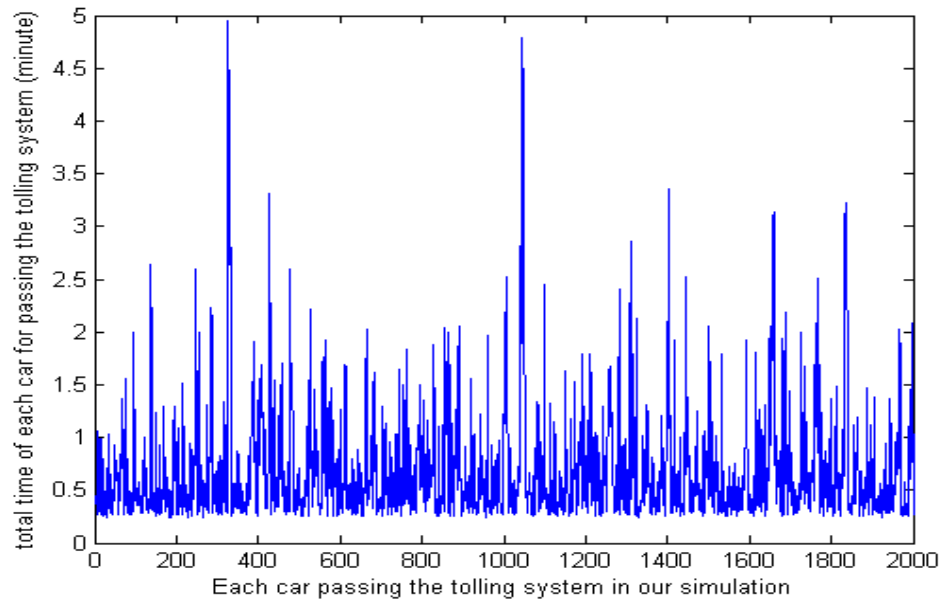


Figure 9. The random pass-time

When we draw the histogram of the distribution of the pass-time, we can easily see the distribution is also an exponential distribution. In fact, in a balance state, we can infer that equation $F_{in} = F_{out}$.

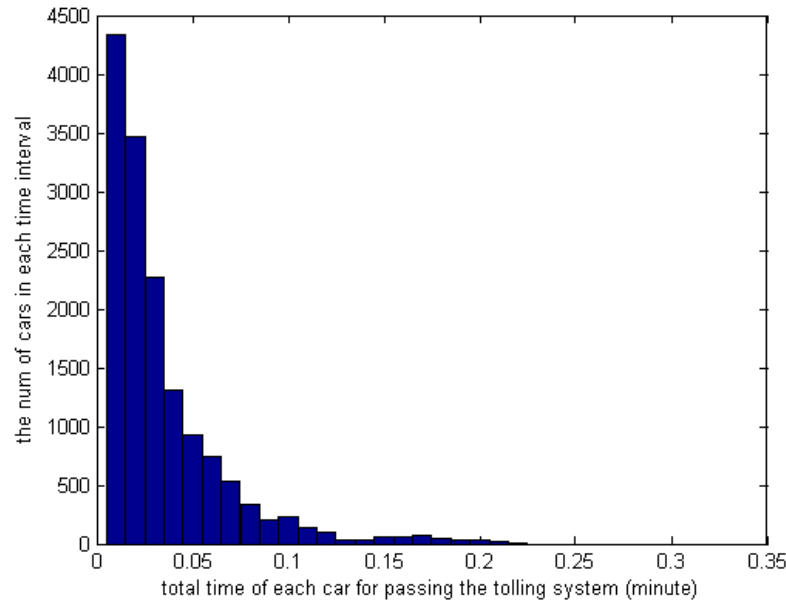


Figure 10. The statistic distribution of the pass-time

When traffic is heavy (for example, in the table, $\lambda=800$ cars/hour), the flow reduces to just 339.6 cars/hour and the pass time also increases to infinity. As we see, the time to pass the tollbooth is still low; by contrast, the time to squeeze into travel lanes is nearly infinity. So, it's easy to explain why congestion increases upon departure from the toll plaza when traffic is heavy.

Here is the diagram to show how the congestion builds: if a congestion is build, the queue will become longer and longer and the pass-time will also increase to infinity.

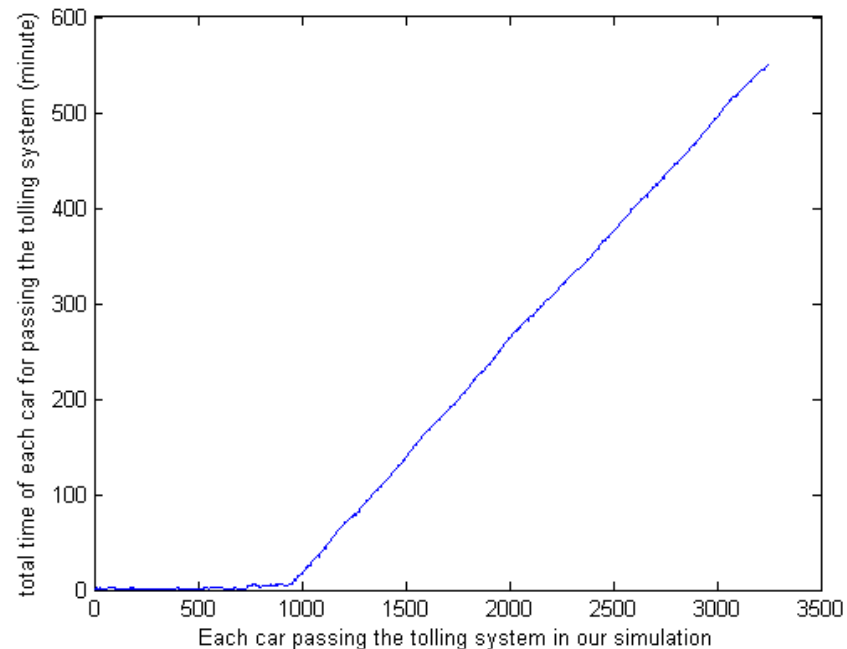


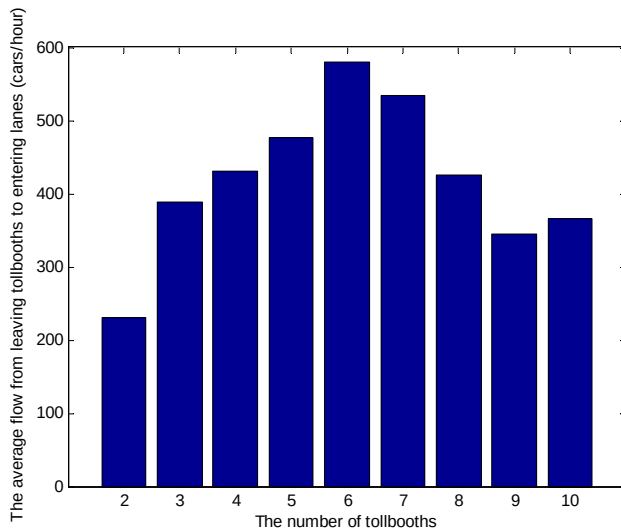
Figure 11. The random pass-time in the situation of congestion

When the traffic is very heavy (for example, in the table, $\lambda=3000\text{cars/hour}$), neither the time to pass tollbooth nor the time of squeezing into lanes will all increase to infinity. That's because the tollbooths have a total maximum average service rate, if the arrival rate is larger than the total maximum average service rate, there will be congestion built at the entry to the toll plaza.

Find the optimal number of tollbooths

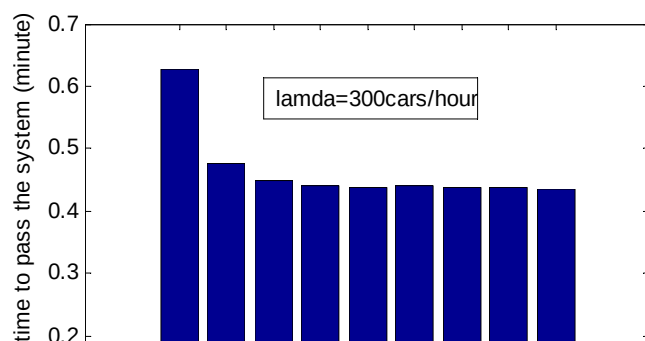
1) Paying by cash or ticket as the Service type

To find the optimal number of tollbooths, we need to compare the pass-time of systems with different number of tollbooths. We continue to consider with the same parameters: $\bar{S}=15\text{s}$, $\bar{L}=50\text{m}$, $\bar{a}=2\text{m/s}^2$. We need to solve the bottleneck in the entry of tollbooths, which means we should make the capability of tollbooths equal to or larger than the capability of squeezing out process. From the table2, we can see the capability of squeezing out process is no more than 700 cars/hour (because there is congestion in the squeezing out process). We search the number of tollbooths that can make the flow maximum in a heavy traffic situation. ($\lambda=650\text{cars/hour}$)



It's obvious to see that 6 tollbooths can make the traffic flow maximum. Because although increase the number of tollbooths can increase the total service capability of tollbooths, too many outgoing cars from tollbooths will increase the density of the outgoing flow, which will make leaving time longer and outgoing flow smaller.

Figure 12. The traffic flows when traffic is heavy



We also need to analyze time to pass the system in different traffic situation. Here is the diagram of the following performance with different number of tollbooths. The diagram shows the situation that traffic is not heavy. In this situation, we can conclude that the increase of the number of tollbooths can also reduce the pass-time. But as we see, since the number is larger than 4, the pass-time

Figure 13. The average pass-times
in ordinary traffic flow

reduces slightly by the increase of the number of tollbooths. We also know that the more tollbooths we have, the more money we'll spend. So we should choose 6 tollbooths for both traffic flow and pass-time consideration and economic consideration.

In this tolling style, from these results by computation, we can easily conclude that we should deploy more tollbooths than the number of incoming lanes.

2) Paying by ETS as the service type

From our early discussion, congestion first appears out from the toll plaza and the bottleneck of the system is the process of squeezing to enter the out lanes. Here are our simulation results of this situation. First we still consider the flow in a congestion situation. ($\lambda=700$ cars/hour).

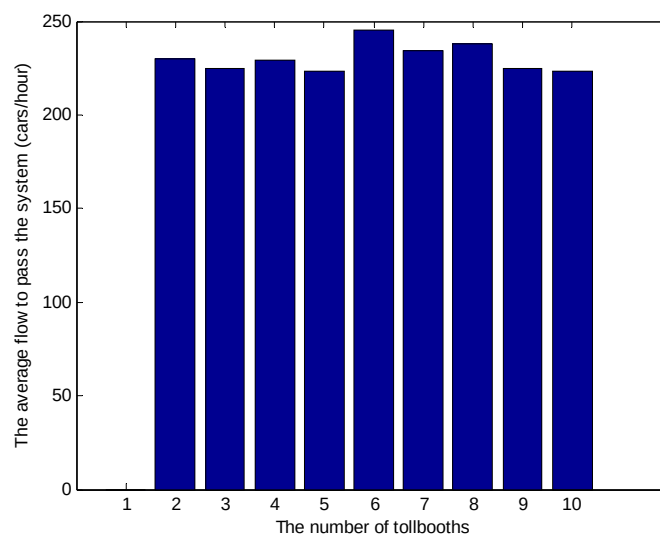
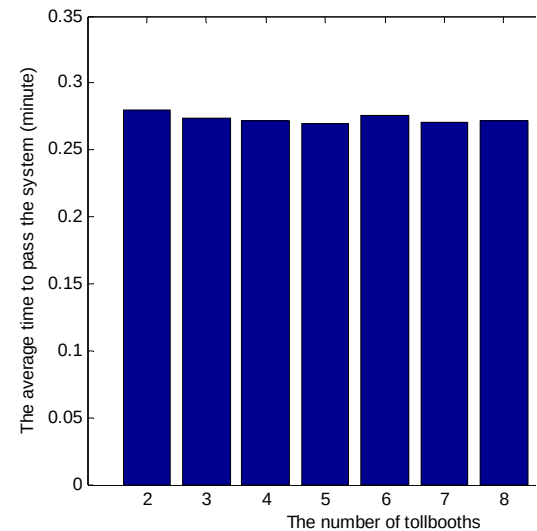


Figure 14. The flows then traffic is heavy

This histogram shows the increase of the number of tollbooths can't improve the traffic flow significantly. That is because the bottleneck of this system isn't the number of tollbooths.

We also analyze the time to pass the system when traffic isn't heavy. As we see from the histogram 2, the pass-time seems nearly unchanged by the increase of the number of tollbooths.



So by our analysis about the results, we can conclude that the increase of tollbooths can't improve the performance of the system. In other words, one tollbooth per incoming travel lane is more effective than current practice because it needs less tollbooths and performs nearly equally to other ones.

Figure 15. Average pass-time in ordinary traffic flow

The effects of different types of vehicles

However, things wouldn't work out as easy as our model. Vehicles are not of the same kind. Consequently, their acceleration parameter and the service time are not the same. To get more reasonable results, we are going to take the vehicles types into account. Here we divide the vehicles into three types: the large-sized, the medium-sized, and the small-sized. Each type has different acceleration parameter and the service time. All the parameters should be empirically obtained. Statistics results in [3] presents the average service time of those three different kinds of vehicles. We currently use the acceleration parameter and the service time with respect to the different type of vehicles as listed in the table 4 below.

Table 4. Vehicles parameters

Vehicle types	Large-sized	Medium-sized	Small-sized
Acceleration (m/s^2)	1.8	2.5	3
Service time paying by cash or ticket(s)	58	10	8.1
Service time paying by ETS (s)	7	2	1
Proportion of each vehicle type	0.2	0.3	0.5

Thus, they vary in their deceleration time and service time. But we can still use the same Algorithm to compute the data we need and find the final optimal number of tollbooths that can meet our objectives best. The method and the analysis of optimization is very same to the method we have discussed clearly, we needn't mention it clearly again, so we apply our program and results in detail in the appendix.

Strengths and weaknesses

Strengths

We constructed our model based on both queuing theory and the relationship between flow density and flow velocity. Thus, the model is grounded in theory and research and therefore reliable and expansible.

We also include in our model the different types of vehicles and tolling. By our simulation, we get a lot of useful data, which we then use to explain the common phenomena like congestions at the tollbooths and the squeezing process.

We still propose three different objectives, which we use to weigh whether a determination of the tollbooths number is optimal. With them together, our final optimal number would bear *higher flow*, *less time*, and *more economical* and thus a really optimal solution.

Weaknesses

Because we build a complex model based on stochastic service systems, it makes the model difficult to compute the analytic optimal number of tollbooths. So we use Monte Carlo algorithm to get the numeric optimal solution. Of course, an analytic solution is more precise than a numeric one.

Also, we don't take the weather conditions into account. However, as we know, different weather conditions affect the traffic significantly. Further study should be put on this problem.

References

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- [4] __. Acceleration data. <http://web.mit.edu/esg/8.01d/www/psets/ps01/>
- [5] Nathan H. Gartner, Carroll J. Messer, and Ajay K. Rathi. 1992. Traffic Flow Theory.
- [6] Kai Nagel, Michael Schreckenberg. 1992. A cellular automaton model for freeway traffic. Journal de Physique I.

Appendix

List of appendixes:

1. The results of simulation and computation with the consideration about different types of vehicles
2. The program to simulate and compute the results without the consideration about different types of vehicles (simulation1.m)
3. The program to simulate and compute the results with the consideration about different types of vehicles (simulation2.m)

(Note: these programs can only run in Matlab 7, we use them to compute many situations just by change their parameters)

Appendix 1: The results of simulation and computation with the consideration about different types of vehicles

Here we assume $\bar{L}=50m$ and other parameters as listed in the table below:

Vehicle types	Large-sized	Medium-sized	Small-sized
Acceleration (m/s^2)	1.8	2.5	3
Service time paying by cash or ticket (s)	58	10	8.1
Service time paying by ETS (s)	9	6	4
Proportion of each vehicle type	0.13	0.37	0.5

Results:

- 1) One tollbooth per incoming lane

By our computation, we get the results as listed in table 1 and table 2. They, respectively, show the results by different service types: tolls paid by cash or ticket, or by passing through the electronic toll system (ETS). As in the **Terms and symbols** section, the variables in the tables are

- λ is the arrival rate;
- $\bar{T}$ is the mean pass-time;
- $\bar{T}_{s1}$ is the mean sojourn time of the first service stage, the tollbooths;
- $\bar{T}_{s2}$ is the mean sojourn time of the second service stage, the plaza-outbound lanes;
- F is the mean traffic flow;

- N_1 is mean number of cars in each tolling queue:
- N_2 is mean number of cars in each leaving queue:

Also, the number in bold and underlined in the tables represents there's a corresponding congestion.

Table 1. Service type: tolls paid by cash or ticket

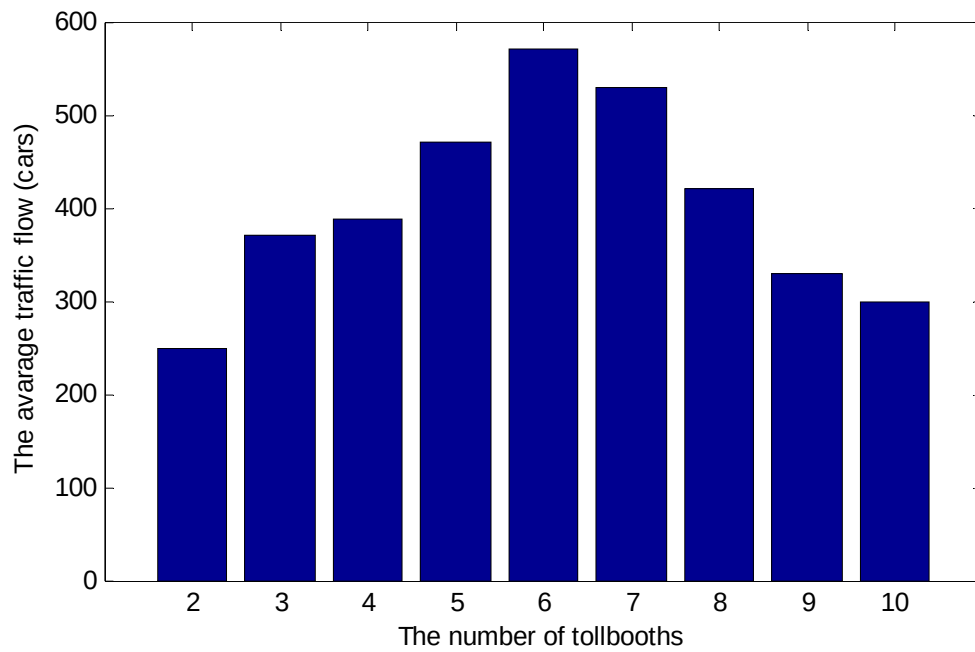
λ (cars/hr)	$\bar{T}$ (s)	$\bar{T}_{s1}$ (s)	$\bar{T}_{s2}$ (s)	F (cars/hr)	N_1 (cars)	N_2 (cars)
300	44.84	38.32	6.62	298.45	2.19	0.79
400	93.36	86.15	7.21	395.15	5.36	0.85
500	<u>2318.7</u>	<u>2311.0</u>	7.68	460.7	<u>162.8</u>	0.93

Table 2. Service type: tolls paid by passing through the ETS

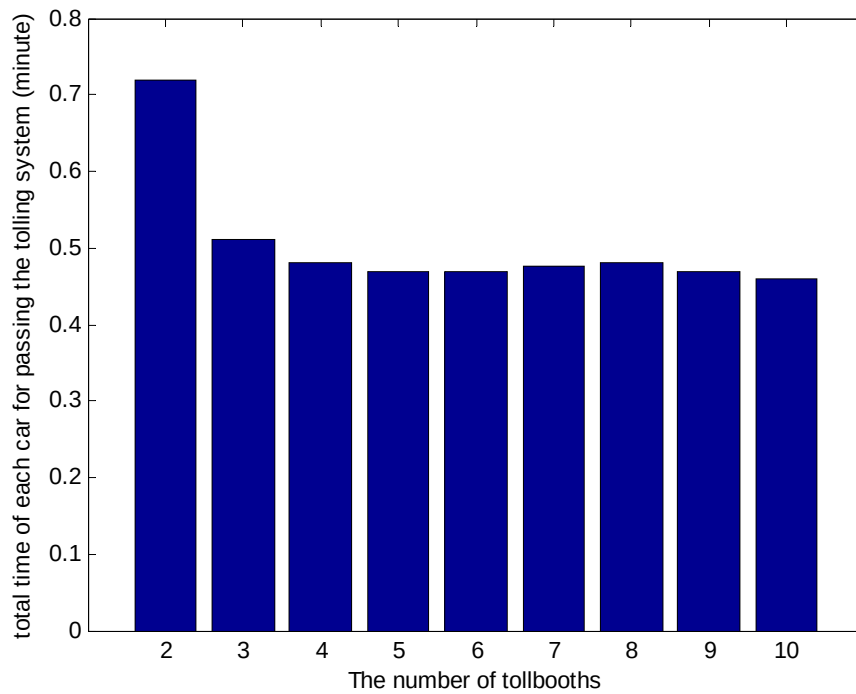
λ (cars/hr)	$\bar{T}$ (s)	$\bar{T}_{s1}$ (s)	$\bar{T}_{s2}$ (s)	F (cars/hr)	N_1 (cars)	N_2 (cars)
400	13.06	6.32	6.74	399.2	0.89	0.80
500	13.96	6.85	7.11	498.3	0.96	0.86
600	14.85	7.26	7.59	596.9	1.06	0.95
700	16.69	8.12	8.57	697.3	1.23	1.13
800	<u>18966</u>	9.53	<u>18956</u>	358.7	1.49	<u>2159</u>
3000	<u>7836.5</u>	<u>4965.0</u>	<u>6491.1</u>	88.2	<u>2072.4</u>	<u>1233.4</u>

2) Find the optimal number of tollbooths

When traffic is heavy ($\lambda=650$ cars/hour).



When traffic isn't heavy ($\lambda=300$ cars/hour)



Appendix 2: Simulation1.m

clear

```

lamda=1/450;u=5/3600;a=2;l=100;n=2;p=1;k=2
% lamda is the average interval time of a car entering the plazas
% k is the number of tollbooths
% u is the average time of service of each type of vehicles
% l is the safety length
% n is the number of lanes
% p is the percentage of each type of vehicles
v=34*1.6/3.6;
car_num=0;
event_num=1;
maxtime=20;
current=0;% the present time
queue_num=zeros(1,k);% the number of cars in every queue waiting for tolling
queue_out=zeros(1,n);% the number of cars in every queue waiting for leaving
time_service=Inf(1,k);% the time a car in a queue paying toll lasts
time_out=Inf(1,n);% the time a car leaves the plaza and enters a lane
qmax=20;
%%%%%%%%%%%%
comenum=0;
qnum=0;
outnum=0;
t1_in=0;
t1_out=0;
t2_in=0;
t2_out=0;
%%%%%%%%%%%%
while (current<=maxtime)
    vihicle_type=rand;
    if vihicle_type<=p(1)
        vihicle_type=1;
    elseif vihicle_type<=p(1)+p(2)
        vihicle_type=2;
    else
        vihicle_type=3;
    end
    t2=sqrt(2*l/a)/3600;
%%%%%%%%%%%%
%%%%%%%%%%%%
% Event-handle module
switch event_num % event_num represents the types of events:1=a car enters the % plazas;2=service(paying
                    toll);3=a car leaves
    case 1
        [m,min_index]=min(queue_num);% find the shortest queue and add in

```

```

if queue_num(min_index)==0
    time_service(min_index)=current+exprnd(u(vehicle_type));
end
% if a queue is null,service the adding car at once.
queue_num(min_index)=queue_num(min_index)+1;
come_time=current+exprnd(lamda);% the time next car comes
%%%%%%%%%%%%%%
comenum=comenum+1;
qnum=qnum+1;
t1_in=t1_in+1;
t_in1(t1_in)=current;
c_t(comenum)=current;
q_n(qnum)=max(queue_num);
q_t(qnum)=current;
%%%%%%%%%%%%%%
case 2
queue_num(index)=queue_num(index)-1;% finish tolling, remove a
                                % car from its queue
if queue_num(index)~=0
    time_service(index)=current+exprnd(u(vehicle_type));
else
    time_service(index)=Inf;
end
[m,min_index]=min(queue_out);% find the shortest out queue and add % in
queue_out(min_index)=queue_out(min_index)+1;
if queue_out(min_index)==1
    time_out(min_index)=current+t2;
end
%%%%%%%%%%%%%%
qnum=qnum+1;
q_n(qnum)=max(queue_num);
q_t(qnum)=current;
t1_out=t1_out+1;
t_out1(t1_out)=current;
%%%%%%%%%%%%%%
t2_in=t2_in+1;
t_in2(t2_in)=current;
outnum=outnum+1;
o_n(outnum)=max(queue_out);
o_t(outnum)=current;
%%%%%%%%%%%%%%
case 3
queue_out(index)=queue_out(index)-1;% a car enters the lane, remove

```

```

% it from its queue
    if queue_out(index)~=0
        k_out=1-(queue_out(index)-1)/qmax;
        if k_out<0.3
            k_out=0.3;
        end
        time_out(index)=current+t2/k_out;
    else
        time_out(index)=Inf;
    end
    %%%%%%%%%%%
    car_num=car_num+1;
    time(car_num)=current;
    %%%%%%%%%%%
    outnum=outnum+1;
    o_n(outnum)=max(queue_out);
    o_t(outnum)=current;
    t2_out=t2_out+1;
    t_out2(t2_out)=current;
    %%%%%%%%%%%
end
%%%%%%%%%%
%%%%%%%%%%

%%%%%%%%%%
%%%%%%%%%%

% choose next event(the event happens earliest)
[sm,si]=min(time_service);% find the next service event
[om,oi]=min(time_out);% find the next out event
[current,event_num]=min([come_time,sm,om]);
switch event_num
    case 2
        index=si;
    case 3
        index=oi;
end
end
%%%%%%%%%%
cx=length(time);
flow(k-1)=car_num;
T_delay=time-c_t(1:cx);
T1_delay=t_out1-t_in1(1:t1_out);
T2_delay=t_out2-t_in2(1:t2_out);

```

```

m_T_delay(k-1)=mean(T_delay);
d_T_delay(k-1)=std(T_delay);
m_T1(k-1)=mean(T1_delay);
m_T2(k-1)=mean(T2_delay);
AV_queue(k-1)=sum(q_n.*[0,diff(q_t)])./q_t(qnum);
AV_outqueue(k-1)=sum(o_n.*[0,diff(o_t)].)/o_t(outnum);

```

Appendix 3: Simulation2.m

```

clear
k=2;lamda=1/100;u=[8.1,10.47,58]/3600;a=[3,2.5,1.8];l=50;n=2;p=[0.5,0.3,0.2];
% lamda is the average interval time of a car entering the plazas
% k is the number of tollbooths
% u is the average time of service of each type of vehicles
% l is the safty length
% n is the number of lanes
% p is the percentage of each type of vihicles
v=34*1.6/3.6;
car_num=0;
event_num=1;
maxtime=20;
current=0;% the present time
queue_num=zeros(1,k);% the number of cars in every queue waiting for tolling
queue_out=zeros(1,n);% the number of cars in every queue waiting for leaving
time_service=Inf(1,k);% the time a car in a queue paying toll lasts
time_out=Inf(1,n);% the time a car leaves the plaza and enters a lane
qmax=20;
%%%%%%%%%%%%
comenum=0;
qnum=0;
outnum=0;
t1_in=0;
t1_out=0;
t2_in=0;
t2_out=0;
%%%%%%%%%%%%
while (current<=maxtime)
    vihicle_type=rand;
    if vihicle_type<=p(1)
        vihicle_type=1;
    elseif vihicle_type<=p(1)+p(2)
        vihicle_type=2;
    else
        vihicle_type=3;
    end
end

```

```

end
t2=sqrt(2*l/a)/3600;
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Event-handle module
switch event_num % event_num represents the types of events: 1=a car enters
                  % the plazas; 2=service(paying toll); 3=a car leaves
case 1
    [m,min_index]=min(queue_num);% find the shortest queue and add in
    if queue_num(min_index)==0
        time_service(min_index)=current+exprnd(u(vehicle_type));
    end
    % if a queue is null,service the adding car at once.
    queue_num(min_index)=queue_num(min_index)+1;
    come_time=current+exprnd(lamda);% the time next car comes
    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
    comenum=comenum+1;
    qnum=qnum+1;
    t1_in=t1_in+1;
    t_in1(t1_in)=current;
    c_t(comenum)=current;
    q_n(qnum)=max(queue_num);
    q_t(qnum)=current;
    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
case 2
    queue_num(index)=queue_num(index)-1;% finish tolling, remove a %car from its queue
    if queue_num(index)~=0
        time_service(index)=current+exprnd(u(vehicle_type));
    else
        time_service(index)=Inf;
    end
    [m,min_index]=min(queue_out);% find the shortest out queue and add % in
    queue_out(min_index)=queue_out(min_index)+1;
    if queue_out(min_index)==1
        time_out(min_index)=current+t2;
    end
    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
    qnum=qnum+1;
    q_n(qnum)=max(queue_num);
    q_t(qnum)=current;
    t1_out=t1_out+1;
    t_out1(t1_out)=current;
    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

```

```

    t2_in=t2_in+1;
    t_in2(t2_in)=current;
    outnum=outnum+1;
    o_n(outnum)=max(queue_out);
    o_t(outnum)=current;
    %%%%%%%%%%%
case 3
    queue_out(index)=queue_out(index)-1;% a car enters the lane, remove %it from its queue
    if queue_out(index)~=0
        k_out=1-(queue_out(index)-1)/qmax;
        if k_out<0.3
            k_out=0.3;
        end
        time_out(index)=current+t2/k_out;
    else
        time_out(index)=Inf;
    end
    %%%%%%%%%%%
    car_num=car_num+1;
    time(car_num)=current;
    %%%%%%%%%%%
    outnum=outnum+1;
    o_n(outnum)=max(queue_out);
    o_t(outnum)=current;
    t2_out=t2_out+1;
    t_out2(t2_out)=current;
    %%%%%%%%%%%
end
%%%%%%%%%%
%%%%%%%%%%

%%%%%%%%%%
%%%%%%%%%%

% choose next event(the event happens earliest)
[sm,si]=min(time_service);% find the next service event
[om,oi]=min(time_out);% find the next out event
[current,event_num]=min([come_time,sm,om]);
switch event_num
    case 2
        index=si;
    case 3
        index=oi;
end

```

```
end
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
cx=length(time);
flow(k-1)=car_num;
T_delay=time-c_t(1:cx);
T1_delay=t_out1-t_in1(1:t1_out);
T2_delay=t_out2-t_in2(1:t2_out);
m_T_delay(k-1)=mean(T_delay);
d_T_delay(k-1)=std(T_delay);
m_T1(k-1)=mean(T1_delay);
m_T2(k-1)=mean(T2_delay);
AV_queue(k-1)=sum(q_n.*[0,diff(q_t)])./q_t(qnum);
AV_outqueue(k-1)=sum(o_n.*[0,diff(o_t)])./o_t(outnum);
```